

# If The God of Technology Made A Wish-List for New Year...

This is what it would look like. These innovations not just demonstrate brilliance in science and technology but also a social spin and street-smart attitude

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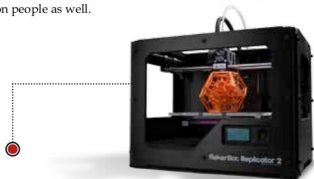
Innovations of a year are worth watching not because they speak of the year gone by, but because they indicate what is in store in the coming year. That is what we had in mind when we chose this set of marvellous innovations from year 2012, which range from motion-activated screwdrivers and 3D printers to social hearing-aids and self-regulating tyres. These innovations demonstrate not just brain power but also street-smart attitude—and they promise to fulfil long harboured desires of not just high-end consumers and techies but common people as well.



## World's first motion-activated screwdriver

Black & Decker has used simple electronics to create a niche product. The 4V Max Gyro is a rechargeable motion-sensing screwdriver that can be controlled by your wrist movement. The device senses the movement of your wrists and uses that as a clue to adjust direction and speed of drilling. You need to rotate your wrist just one quarter turn to the right for forward, or left for reverse.

**Inside:** Its Inverse ISZ-650 Z-axis integrated micro-electromechanical-system (MEMS) gyroscope is designed for high performance, robustness and high shock resistance. MEMS technology makes it small enough for a compact screwdriver. Light-emitting diode (LED) lights are used to illuminate the workspace; plus an effective lithium-ion battery ensures that the device remains charged for as long as one-and-a-half year.



## A factory in your garage

The MakerBot Replicator 2 is a compact desktop device that lets you print 3D objects. It works using Fused Filament Fabrication (FFF) technology. Basically, it melts plastic and squeezes it through an extruder to build an object layer-by-layer. Replicator 2 uses polylactide filament—a super-strong renewable plastic that can be used to make even movable parts and complex assemblies. The device allows print dimensions of  $28.5 \times 15.3 \times 15.5 \text{ cm}^3$ , with a path-breaking 100-micron resolution ensuring smooth surfaces and fine features.

**Inside:** It uses modified version of Mighty Board—MakerBot's original 3D printer control technology—that can control up to five stepper motors with BotStep stepper controllers, two extruders (with heater cartridges, safety thermostats, thermocouples and fans), Replicator interface board and heated build platform, apart from connecting up to six end-stops. The extruder is nearly identical to the original Replicator extruder with minor tweaks to optimise manufacturing of injection moulded parts. Other features include 3-point levelling system and 32 per cent lower power consumption.

While the original Replicator design was completely open, there is a lot of furore about the company moving to a closed model now. However, the founder claims that the core technology comprising the extruder and the control board are still open. Original component information is available at <http://www.makerbot.com/blog/tag/mightyboard>